

HYDRA: Explainability-Driven Intrusion Detection for IoT Network-Flow Data

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Abstract

Machine-learning-based Intrusion Detection Systems (IDS) achieve high benchmark performance yet frequently operate as opaque black boxes, issuing alerts without reliable justification. This opacity creates direct operational costs: analysts must investigate alerts without trustworthy guidance, false positives generate alert fatigue, and trust in automated detection degrades even when headline accuracy is strong.

This dissertation presents HYDRA, an explainability-driven IDS framework that treats model explanations as measurable system outputs. HYDRA addresses two tasks simultaneously: binary attack detection and multi-class attack-type classification, reflecting the operational need to know both that an anomaly has occurred and which threat category it represents. The core model set comprises majority-class and threshold baselines, Logistic Regression (as a reference explainer), Random Forest, XGBoost, and LightGBM, evaluated on TON_IoT as the primary benchmark and CIC-IoT-2023 for generalisation. A CNN-LSTM model achieves PR-AUC of 0.9998 on TON_IoT; a Graph Neural Network achieves PR-AUC of 0.9997.

Explanations are generated using a portfolio of post-hoc methods matched to each model family: SHAP TreeExplainer for the tree ensembles, SHAP LinearExplainer for Logistic Regression, and LIME applied model-agnostically across all models. Integrated Gradients is implemented for the CNN-LSTM and GNN extensions. Every explanation method is evaluated against five measurable criteria: faithfulness (comprehensiveness and sufficiency under top- k feature ablation), stability (Spearman rank correlation under noise perturbation and across independent retraining runs), simplicity (k_{90} fraction and Gini coefficient), plausibility (RMA@ k against expert network features), and timeliness (milliseconds per sample).

The contribution is not a novel detection algorithm but a reproducible evaluation framework that operationalises explainability as a measurable system property, enabling rigorous assessment of whether high-performing IDS models can provide explanations that are faithful, stable, and useful enough to support analyst decision-making in practice.

On TON_IoT, all four tabular models achieve PR-AUC of 1.000 and ROC-AUC of 1.000 under the behaviour-only regime, with XGBoost achieving the highest OXS explanation quality score of 0.828. Cross-dataset evaluation on CIC-IoT-2023 reveals that tree ensemble models suffer catastrophic ROC-AUC drops of up to 87%, while Logistic Regression generalises most robustly (ROC-AUC drop of 7.3%). The leakage interaction analysis demonstrates that including identifier features reduces explanation plausibility by 35 percentage points (Cohen’s $d = -2.985$) without affecting detection performance.

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Glossary of Abbreviations

Abbreviation	Definition
AOPC	Area Over the Perturbation Curve
AUC	Area Under the Curve
CNN	Convolutional Neural Network
DoS	Denial of Service
GNN	Graph Neural Network
IDS	Intrusion Detection System
IG	Integrated Gradients
IoT	Internet of Things
LIME	Local Interpretable Model-agnostic Explanations
LSTM	Long Short-Term Memory
OXS	Operational eXplainability Score
PR-AUC	Precision-Recall Area Under the Curve
RF	Random Forest
ROC-AUC	Receiver Operating Characteristic AUC
SHAP	SHapley Additive exPlanations
SOC	Security Operations Centre
XAI	Explainable Artificial Intelligence

Chapter 1

Introduction

1.1 Motivation

The cyber-defence landscape is defined by a fundamental asymmetry: defenders must secure an ever-expanding and heterogeneous attack surface, while adversaries require only a single successful compromise. Automated scanning and exploitation infrastructure operates at tens of thousands of scans per second, and credential-based attacks have risen markedly [21]. Simultaneously, enterprises generate billions of telemetry events per day across servers, endpoints, cloud services, and IoT devices, creating a signal-to-noise problem that overwhelms traditional monitoring and human analysts [15].

Conventional controls—signature-based IDS (Snort, Suricata) and rule-driven SIEM platforms—are fundamentally reactive: they cannot detect novel attacks, polymorphic malware, or living-off-the-land techniques [2], and produce high false positive rates as system complexity grows [67].

AIOps for cybersecurity seeks to address these limitations by embedding machine learning into the Security Operations Centre (SOC). Industry analyses suggest alert noise reductions of 70–90% [43]. However, learned models introduce new operational challenges: deep learning systems behave as opaque black boxes, real-time inference imposes strict latency constraints, and models are vulnerable to concept drift and adversarial manipulation [57, 23].

1.2 Problem Statement

A central failing of current IDS research is the gap between detection performance and operational usefulness. High benchmark accuracy on public datasets does not translate directly into deployable, trustworthy systems for three interconnected reasons.

First, many published results are inflated by evaluation leakage. When network-flow records are split randomly by row, similar traffic from the same hosts appears in both training and test sets, allowing models to recognise identity patterns rather than attack behaviour [64]. This leakage is invisible under standard evaluation but causes catastrophic degradation in deployment.

Second, most IDS models provide binary outputs without indicating the threat category. In practice, different attack types—denial-of-service, ransomware, exfiltration, scanning—require different analyst responses. Dual-task detection and classification

is substantially more useful operationally.

Third, accurate models are underutilised because alerts lack justification. When a model flags a flow as malicious without explaining which features triggered the decision, analysts must investigate blindly, increasing triage time and eroding trust in automation.

1.3 Aim and Objectives

The aim of this project is to design, implement, and evaluate HYDRA: an intrusion detection framework that addresses both binary attack detection and multi-class attack-type classification, while generating per-alert explanations that are systematically validated for reliability, consistency, and operational usefulness on IoT network-flow data.

The specific objectives are:

1. To implement and evaluate a structured model progression—from baselines through tabular ensemble models and deep learning architectures—for both binary and multi-class intrusion detection on TON_IoT, with CIC-IoT-2023 as a secondary generalisation benchmark. A CNN-LSTM model and a Graph Neural Network (GraphSAGE with Integrated Gradients attribution) are evaluated as deep learning extensions.
2. To define and apply two feature regimes that explicitly isolate and quantify the contribution of identifier leakage to detection performance and explanation quality.
3. To generate standardised per-alert explanation records for tabular models using SHAP TreeExplainer, for the Logistic Regression reference model using SHAP LinearExplainer, and for all models using LIME as a model-agnostic cross-reference. Integrated Gradients is implemented for the CNN-LSTM and GNN extensions.
4. To evaluate explanations systematically using five measurable criteria: faithfulness (comprehensiveness and sufficiency), stability (under noise perturbation and across independent retraining seeds), simplicity (k_{90} and Gini), plausibility (RMA@ k), and timeliness (ms per sample).
5. To position HYDRA explicitly against concurrent XAI-IDS work and compare the performance–explainability trade-off across model types, with generalisation evaluation on CIC-IoT-2023 where feasible.

1.4 Research Questions

This project is structured around four research questions:

RQ1 (Detection): Which modelling approach provides the strongest performance for both binary attack detection and multi-class attack-type classification on tabular network-flow features, when evaluated using operationally relevant metrics such as PR-AUC and false positive rate at a fixed recall target?

RQ2 (Explainability): To what extent can the best-performing detector provide per-alert, feature-level explanations with high and consistent coverage across both detection tasks?

RQ3 (Reliability): How stable are explanation outputs across repeated training runs and different data conditions, and does stability differ between the binary and multi-class tasks?

RQ4 (Faithfulness): Do model-generated explanations exhibit fidelity—that is, does removing the highlighted features significantly change model confidence—and do explanation patterns align with known network intrusion behaviours for each attack type?

1.5 Contributions

This project does not propose a novel detection algorithm. Its contributions are:

- A reproducible IDS pipeline that supports both binary and multi-class intrusion detection with standardised per-alert explanation records for tabular ensembles, with CNN-LSTM (PR-AUC = 0.9998) and GNN (PR-AUC = 0.9997) deep learning extensions producing valid detection results and Integrated Gradients attributions.
- A unified, implementation-ready explainability evaluation protocol covering five measurable criteria (faithfulness, stability, simplicity, plausibility, timeliness), applied consistently across all evaluated model families and both detection tasks. Importantly, the stability criterion is operationalised as two separate experiments: seed-retraining stability (training $S = 5$ models with different random seeds) and noise-perturbation robustness (perturbing fixed model inputs), which address distinct aspects of explanation reliability.
- Empirical analysis of the impact of identifier-based feature leakage on detection performance and explanation quality in IoT network-flow datasets, and explicit positioning of HYDRA against concurrent XAI-IDS systems through a gap analysis table.
- A systematic empirical analysis of how evaluation methodology—specifically the choice of data splitting strategy—inflates both detection and classification performance in IoT intrusion detection. Host-disjoint evaluation reduces the best model’s multiclass macro-F1 from 1.000 to 0.742 (Random Forest) and XGBoost’s binary ROC-AUC from 1.000 to 0.842, even when identifier features are excluded, demonstrating that host-level flow correlation is a previously underreported source of evaluation leakage.

To the best of the author’s knowledge, HYDRA is the first system to empirically implement and validate all of the following simultaneously: (i) dual-task binary and multiclass detection evaluated under leakage-controlled feature regimes, (ii) a five-criterion quantitative XAI evaluation protocol producing a composite OXS score, (iii) cross-dataset generalisation testing on IoT-specific benchmarks, and (iv) empirical measurement of the interaction between identifier leakage and explanation quality. While individual components appear in the literature (Table 2.2), their combined application to IoT intrusion detection with measurable explanation evaluation is novel.

Table 1.1: Summary of HYDRA’s contributions and their relation to research gaps.

Contribution	Gap Addressed	Key Evidence
Dual-task IDS pipeline	Binary-only detection (§2.7.2)	Macro-F1 = 1.000 (Ch. 4)
Five-criterion XAI protocol	XAI as afterthought (§2.7.3)	OXS ranking (Table 4.14)
Leakage interaction analysis	Evaluation leakage (§2.7.1)	RMA@5 Δ = -0.35 (Table 4.16)
Cross-dataset generalisation	Single-dataset evaluation	ROC-AUC drop up to 87% (Table 4.15)
Split-strategy leakage analysis	Overoptimistic benchmarks	Macro-F1: 1.000 $\rightarrow$ 0.742 (Table 4.3)

1.6 Dissertation Structure

Chapter 2 presents the literature, technology, and data survey. Chapter 3 describes the HYDRA methodology. Chapter 4 presents experimental results. Chapter 5 discusses findings, trade-offs, and limitations. Chapter 6 concludes and identifies future work.

Chapter 2

Literature, Technology and Data Survey

2.1 Evolution of Intrusion Detection

Intrusion detection has evolved through three generations. Signature-based systems (Snort, Suricata) match packet contents against known Indicators of Compromise, offering high precision for catalogued threats but fundamentally unable to detect novel attacks, polymorphic variants, or adversaries using legitimate credentials [2]. Rule-based SIEM platforms extend this by correlating events across sources, but static rules become brittle as system complexity grows and produce high false positive rates on heterogeneous telemetry [67].

Anomaly-based IDS addressed the zero-day gap by learning statistical baselines of normal behaviour. Supervised ML methods—Random Forest, SVMs, gradient boosting—achieved high accuracy on benchmarks such as KDD Cup 99 and CICIDS2017 [4], though they generalise poorly to unseen attack types and suffer concept drift. Hybrid architectures combining supervised classification with unsupervised anomaly detection improve zero-day recall [5], a principle that underpins HYDRA’s design.

AIOps platforms automate IT security operations through ML-driven noise reduction, correlation, and root-cause analysis [15]. Industry analyses suggest alert noise reductions of 70–90% [21], though commercial platforms are closed and not benchmarked on standardised open datasets. HYDRA provides a transparent, reproducible alternative.

2.2 Supervised and Unsupervised Learning for Cyber Defence

2.2.1 Supervised Ensemble Methods

Ensemble tree methods form the core of HYDRA’s tabular detection pipeline. Random Forest [12] constructs an ensemble of independently trained decision trees via bootstrap sampling with random feature subsets, producing stable feature importance estimates. XGBoost [14] implements gradient boosted trees through second-order gradient approximation and regularisation. LightGBM [28] extends this through histogram-based splitting for memory efficiency. Logistic Regression serves as a reference explainer whose linear decision boundary yields deterministic, auditable attributions.

In the IDS literature, ensemble models consistently achieve state-of-the-art results [45, 51], though many studies use random row-level splits without temporal ordering, raising concerns about evaluation leakage. A key challenge for supervised IDS is class imbalance: standard accuracy metrics are misleading when attack events represent a small fraction of total traffic, making precision-recall metrics and PR-AUC essential for operational evaluation.

2.2.2 Unsupervised Methods

Isolation Forest [33] isolates anomalies by recursively partitioning the feature space with random cuts; anomalous points require fewer partitions. It operates in approximately linear time, making it practical as a fast pre-filter [5]. Autoencoders trained on normal traffic detect anomalies via high reconstruction error [40]; LSTM-autoencoders extend this to sequential data [38?]. HYDRA does not use these methods directly but they inform its design philosophy.

2.3 Deep Learning for Log and Network-Flow Analysis

DeepLog [17] treats log entries as sequences of discrete keys and trains an LSTM to predict the next expected key, flagging deviations as anomalies. LogBERT [25] adapts BERT’s masked prediction to log sequences, conditioning on both left and right context for improved detection. Transformer-based models achieve higher accuracy than LSTM baselines on LogHub benchmarks but at substantially higher computational cost [25?].

For network flow data, hybrid CNN-LSTM architectures capture both spatial (feature interaction) and temporal (sequence) structure simultaneously. Studies on TON_IoT demonstrate detection accuracy above 99% for binary classification [? 26]. Federated variants [48] keep raw traffic local to each device, addressing privacy constraints in distributed IoT environments.

2.4 Graph-Based Detection and Predictive Analytics

Network communications have natural graph structure: hosts are nodes and flows are edges. GNNs exploit this relational structure through message-passing operations, learning representations that capture communication topology—particularly relevant for detecting scanning, lateral movement, and botnet coordination. Euler [29] uses temporal link prediction and LMTracker [34] employs heterogeneous graph embedding for lateral movement detection on authentication graphs.

Predictive analytics extends detection to forecasting. LSTM-based approaches forecast DDoS intensity [30], and sequence models on authentication patterns predict lateral movement by flagging low-probability authentications [64]. A survey of data-driven methods [59] identifies evaluation metrics, dataset selection, and reliability under concept drift as primary challenges.

2.5 Explainability in Intrusion Detection

2.5.1 The Explainability Problem in IDS

High-performing ML models used for intrusion detection frequently operate as black boxes, producing alerts without exposing the features or reasoning behind them. This increases triage time, compounds alert fatigue, and erodes the trust required for automation in high-stakes SOC environments. Post-hoc XAI methods explain already-trained models without modifying them, and are more widely used than intrinsically interpretable models because they can be applied to high-performing ensembles [1].

HYDRA implements four complementary post-hoc methods: SHAP TreeExplainer for tree ensembles, SHAP LinearExplainer for Logistic Regression, LIME as a model-agnostic cross-reference, and Integrated Gradients for the CNN-LSTM and GNN.

2.5.2 SHAP TreeExplainer

SHAP [35] grounds feature attribution in cooperative game theory, computing each feature’s Shapley value—its average marginal contribution across all possible feature orderings. TreeExplainer [36] provides an exact, polynomial-time implementation for tree-based models, avoiding the sampling approximation of model-agnostic variants. In HYDRA, TreeExplainer produces both local per-alert attributions and global feature importance rankings. SHAP values can be unstable under correlated features, which HYDRA addresses through explicit stability measurement.

2.5.3 SHAP LinearExplainer

For Logistic Regression, LinearExplainer computes Shapley values analytically by exploiting the additive structure of linear prediction. Because attributions are deterministic given the same data, LogReg serves as a stability anchor for comparing against tree ensemble models, consistent with the performance–explainability trade-off analysis of [10] and [1].

2.5.4 LIME

LIME [53] explains individual predictions by fitting a locally linear surrogate model to perturbed versions of the input. HYDRA applies LIME across all models as a model-agnostic cross-reference against SHAP attributions. LIME’s primary limitation is sampling variance; comparative studies find SHAP more stable for tabular data [65], so HYDRA treats LIME as a secondary method.

2.5.5 Integrated Gradients

Integrated Gradients [62] computes feature attributions by integrating the model’s gradients along a path from a baseline input to the actual input. It satisfies sensitivity and implementation invariance axioms. In HYDRA, IG is applied to both the CNN-LSTM and the GNN’s edge feature space.

2.5.6 GNNExplainer and Policy Graphs

GNNExplainer [66] learns a soft mask over a node’s computational subgraph that maximises mutual information with the full-graph prediction. Subgraph explanations can be non-unique and may be dominated by topology rather than feature content. Policy graphs [7] and provenance graphs [18] represent access control and causal event chains, informing the plausibility criterion in HYDRA’s evaluation.

2.5.7 XAI Evaluation: Five Criteria

A recurring weakness in XAI-IDS research is the absence of systematic evaluation of explanations. HYDRA evaluates every method against five criteria (Table 2.1).

Table 2.1: HYDRA’s five-criterion XAI evaluation framework.

Criterion	Metric	Description
Faithfulness	Comprehensiveness + Sufficiency (top- k ablation)	Remove top- k features (comprehensiveness) or keep only top- k (sufficiency) and measure confidence change.
Stability	S1: Spearman ρ across $S = 5$ re-training runs; S2: Spearman ρ under Gaussian noise	Two distinct experiments addressing training stochasticity (RQ3) and input perturbation (RQ4).
Simplicity	k_{90} fraction + Gini coefficient	Minimum features for 90% attribution mass (normalised); concentration of attribution distribution.
Plausibility	RMA@ k vs expert features	Proportion of top- k attributed features in an expert-defined reference set.
Timeliness	ms per sample	Wall-clock time for a complete explanation record.

Faithfulness is measured through top- k ablation: comprehensiveness removes the top- k features and measures confidence drop; sufficiency retains only top- k and measures whether they reproduce the original confidence [10, 16]. Stability uses Spearman rank correlation, which captures the full ordering of feature importance and is robust to ties. Simplicity combines k_{90} (fraction of features needed for 90% of attribution mass) with the Gini coefficient of the attribution distribution. Plausibility compares top- k features against expert-defined reference sets for each attack type. Timeliness constrains which methods are viable for real-time deployment.

2.6 Positioning HYDRA Against Concurrent Work

Four recent papers address XAI-IDS in settings sufficiently similar to HYDRA that explicit positioning is necessary. Table 2.2 maps each against HYDRA’s principal design choices.

Table 2.2: Positioning HYDRA against concurrent XAI-IDS work. ✓ = present, ○ = partial, × = absent.

Dimension	HYDRA	E-XAI [27]	XAI-IDS [60]	Frontiers [11]	TalTech [65]
Binary + multi-class detection	✓	×	×	×	×
Leakage-controlled feature regimes	✓	×	×	×	×
Temporal (non-random) splitting	✓	×	×	○	×
≥3 XAI methods compared	✓	✓	✓	○	✓
Stability measured quantitatively	✓	×	×	×	○
Timeliness (ms/sample) measured	✓	×	×	×	×
Simplicity (k_{90} , Gini) measured	✓	×	×	×	×
Plausibility (RMA@ k) measured	✓	×	×	×	×
IoT-specific dataset (TON_IoT)	✓	×	○	✓	×
Cross-dataset generalisation	✓	×	×	×	×

E-XAI [27] evaluates SHAP, LIME, and IG on deep-learning NIDS but addresses binary classification only, without leakage-controlled features, quantitative stability, or cross-dataset generalisation. XAI-IDS [60] applies SHAP and LIME to CICIDS2017 with random splitting and no quantitative XAI evaluation. The Frontiers study [11] applies SHAP, LIME, and ELI5 but is primarily qualitative. The TalTech study [65] compares four XAI methods on real SOC data using a four-criterion framework, finding DeepLIFT superior to SHAP; HYDRA’s addition of timeliness and simplicity criteria, IoT-specific focus, and multi-class evaluation are key differentiators.

Two further works warrant acknowledgement. [1] propose a conceptual framework for the tension between detection, efficiency, and explanation quality, but without empirical implementation. [58] introduce a domain-agnostic explanation reliability index but do not address IDS-specific evaluation. HYDRA is the first system to implement and validate all ten dimensions simultaneously.

2.7 Gaps in Current Technology

2.7.1 Evaluation Leakage and Overoptimistic Benchmarks

A pervasive problem in IDS research is evaluation leakage that inflates performance. Random row-level splitting places correlated flows from the same hosts or time windows into both training and test sets, allowing models to recognise identity patterns rather than attack behaviours [31]. Including identifier features (IPs, ports, URIs) allows models to memorise which sources were associated with attacks [19]. HYDRA addresses both through deployment-realistic splits and explicit feature regime comparison.

2.7.2 Binary-Only Detection

Most published IDS models evaluate solely binary classification despite operational response differing substantially by threat type. HYDRA’s dual-task design addresses this directly.

2.7.3 Explainability as a Qualitative Afterthought

Even IDS studies incorporating explainability present explanations as static plots without measuring stability, fidelity, or coverage. HYDRA's five-criterion framework addresses this gap.

2.7.4 Adversarial Robustness and Concept Drift

Security ML models operate in adversarial environments where attackers may attempt evasion or data poisoning [57, 23], and concept drift renders models obsolete without retraining [22]. HYDRA's modular architecture supports periodic re-evaluation and is designed for extension with drift detection.

2.8 Technology Survey

HYDRA uses scikit-learn [47] for classical ML and preprocessing, XGBoost and LightGBM for gradient-boosted trees, PyTorch [46] with PyTorch Geometric [20] for the GNN, SHAP [37] for Shapley values, LIME for local surrogate explanations, and Capsum for neural network attribution. Apache Kafka [9] provides streaming infrastructure [52, 39]; Grafana [24] and Streamlit [61] provide visualisation and interactive analyst workbench capabilities.

2.9 Data Survey

2.9.1 TON_IoT (Primary Dataset)

The TON_IoT dataset [40], from UNSW Canberra, contains millions of labelled flow records from a realistic IoT testbed spanning nine attack categories: Scanning, DoS, DDoS, Injection, Backdoor, XSS, Ransomware, Password, and MitM. Each record contains binary and categorical labels supporting dual-task evaluation. Known challenges include class imbalance, synthetic attack artefacts, and label leakage through identifier features [44?].

2.9.2 CIC-IoT-2023 (Generalisation Dataset)

CIC-IoT-2023 [13] contains approximately 3.18 million flow records after deduplication (44% duplicates removed), with 34 attack types and 97.7% attack prevalence. It lacks source IP and timestamp columns, constraining splitting strategies and precluding GNN evaluation. Its greater attack diversity makes it a useful generalisation testbed.

2.9.3 Supporting Datasets

LogHub [67] aggregates labelled log datasets (HDFS, BGL, Thunderbird) used as benchmarks for log anomaly detection methods. The LANL dataset [64] contains billions of de-identified authentication events from an enterprise network with embedded Red Team activities, informing HYDRA's graph construction approach.

2.9.4 Data Preparation

HYDRA applies consistent preprocessing: feature extraction derives numerical features and protocol flags, class imbalance is addressed through stratified sampling and class-weighted loss functions [8], temporal splitting respects time-ordering where timestamps are available, and all preprocessing is fitted exclusively on training data.

2.10 Summary

The literature establishes three conclusions motivating HYDRA. First, ensemble tree methods dominate tabular IDS benchmarks, yet results are frequently inflated by identifier leakage and random splitting, motivating HYDRA’s dual-regime evaluation. Second, SHAP provides more stable attributions than LIME for tree models, but LIME’s model-agnostic nature makes it the only viable cross-reference across model families, motivating HYDRA’s multi-method portfolio. Third, no existing study applies a unified quantitative XAI evaluation protocol across multiple model families on both binary and multi-class IoT detection under leakage-controlled conditions—the primary gap HYDRA addresses.

Chapter 3

Methodology

3.1 Overview

HYDRA implements a two-stage classification pipeline: Stage 1 performs binary detection (attack vs. normal); Stage 2 classifies detected attacks by type. The experimental design spans three dimensions: **detection performance** across five ML models and two deep learning architectures under two feature regimes; **explanation quality** assessed via a five-criterion protocol; and **cross-dataset generalisation** testing portability of learned representations.

All experiments use TON_IoT and CIC-IoT-2023. Figure 3.1 illustrates the pipeline architecture.

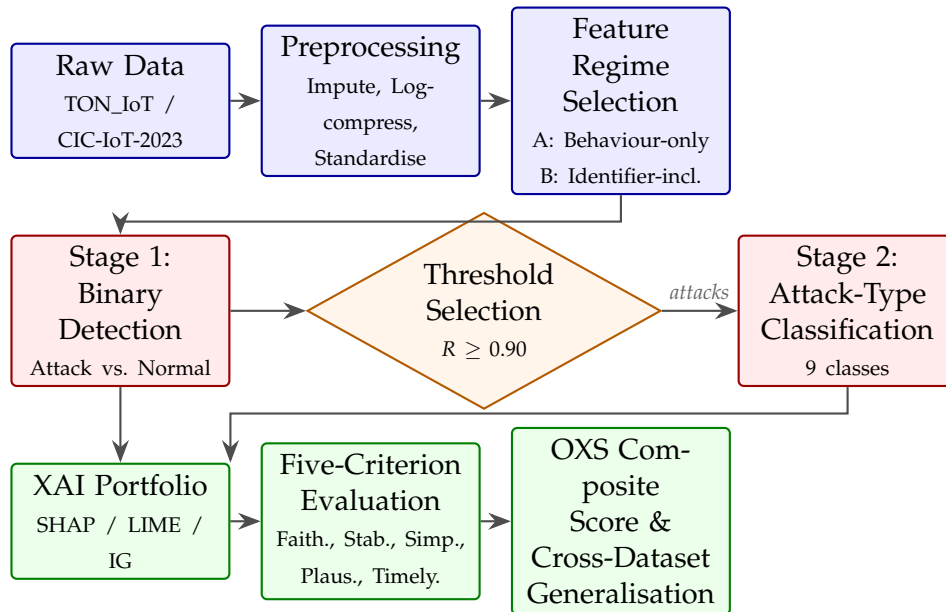


Figure 3.1: HYDRA pipeline architecture. Raw data flows through preprocessing and feature regime selection into the two-stage detection pipeline. Both stages feed into the XAI portfolio, assessed against five criteria and aggregated into the OXS composite score.

3.2 Datasets and Preprocessing

3.2.1 TON_IoT

TON_IoT [40] comprises approximately 211,000 bidirectional network flow records extracted using the Zeek network monitor. Each row is characterised by connection meta-data (source/destination IP, port, protocol), volumetric statistics (bytes and packets in each direction), and temporal features (duration, timestamps). The dataset contains nine attack types—DoS, DDoS, scanning, ransomware, backdoor, injection, XSS, password, and MitM—alongside normal traffic, with 76% attack prevalence. Source IP addresses enable group-disjoint splitting.

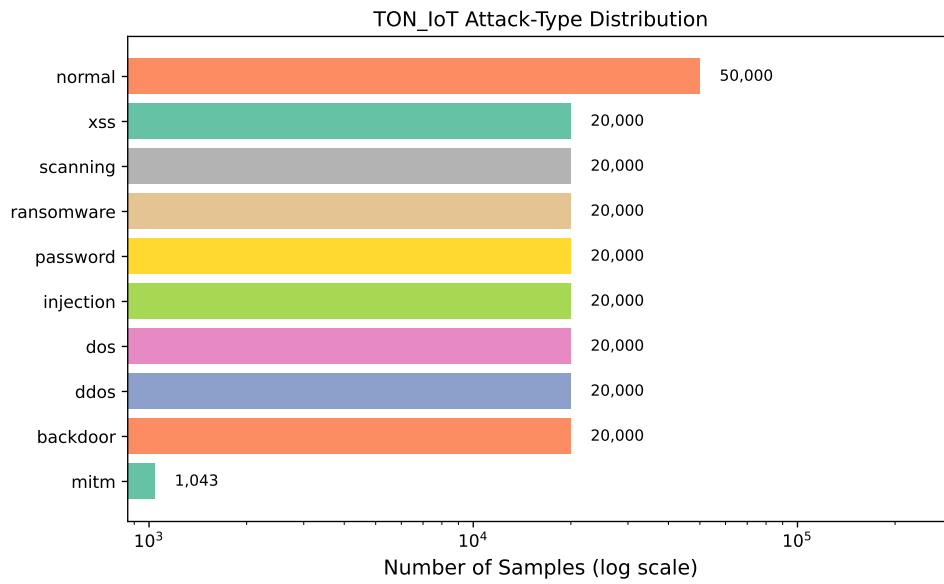


Figure 3.2: Class distribution of TON_IoT dataset (log scale).

3.2.2 CIC-IoT-2023

CIC-IoT-2023 [13] comprises approximately 3.18 million flow records after deduplication. The raw dataset contained 44% duplicate rows; a preprocessing step computes SHA-256 hashes and removes exact duplicates to prevent train-test leakage. After deduplication, the dataset contains 34 attack types with 97.7% attack prevalence. Features include protocol flags (TCP, UDP, ICMP), aggregate packet statistics, and statistical summaries. It lacks IP and timestamp columns, constraining splitting strategies and precluding GNN evaluation.

3.2.3 Preprocessing Pipeline

The preprocessing pipeline is fitted exclusively on the training partition to prevent information leakage. It applies three sequential transformations:

1. **Median imputation:** missing numeric values are replaced with the training-set median. Categorical missing values are imputed with the most frequent training-set value.

2. **Log-compression:** a $\log(1 + x)$ transformation is applied to all non-negative numeric features after clipping negative values to zero, compressing the right-skewed distributions characteristic of network flow volumes.
3. **Standardisation:** each numeric feature is centred to zero mean and scaled to unit variance using training-set statistics. While tree-based models are scale-invariant, this step benefits logistic regression and the CNN-LSTM.

Categorical features undergo most-frequent imputation followed by one-hot encoding with unknown-category handling, ensuring that categories unseen during training do not cause errors at inference.

3.3 Feature Regimes

Regime A (Behaviour-only) retains only traffic-behavioural features: flow duration, byte counts (source and destination), packet counts, protocol metadata, and connection state indicators. It explicitly excludes IP addresses, port numbers, service labels, and high-cardinality string fields. By removing identifiers, Regime A forces models to learn from traffic patterns alone, producing explanations that reference only operationally meaningful features.

Regime B (Identifier-inclusive) augments Regime A with network identifiers: source and destination IP addresses, port numbers, and service labels. This permits models to exploit identity-based shortcuts—for example, memorising that a specific source IP is always associated with attacks.

The distinction serves a dual purpose. For detection, it quantifies the performance gap attributable to identifier leakage. For explainability, it tests whether Regime B models produce explanations dominated by identifier features that offer no actionable insight (Appendix A).

3.4 Data Splitting Strategies

HYDRA implements four data splitting strategies, each producing disjoint train/validation/test partitions.

Stratified split uses scikit-learn’s `StratifiedShuffleSplit` to preserve class distribution across all partitions. It is the primary strategy because it guarantees sufficient positive and negative samples in each split.

Host split groups flows by source IP and assigns entire hosts to a single partition, ensuring no IP appears in more than one split. This prevents models from memorising host-specific patterns and simulates deployment against unseen devices. Applicable only to TON_IoT.

Temporal split orders flows chronologically and assigns earliest to training, middle to validation, latest to test. Applicable only to TON_IoT.

Host-type-aware split extends host splitting with a greedy pinning algorithm that guarantees all attack types appear in training by pinning the host with the most samples of each type (rarest first).

All strategies enforce non-empty partitions, no NaN labels, and class-presence warnings. Stratified splitting is primary; host and temporal splits serve as supplementary deployment-realistic analyses.

3.5 Model Selection and Architecture

Two non-learned baselines establish lower bounds. The **majority-class baseline** assigns every flow a score equal to the training-set attack prevalence. The **threshold baseline** computes a heuristic anomaly score as $\log(1 + \text{duration}) + \log(1 + \text{total_bytes}) + \log(1 + \text{total_packets})$, capturing the intuition that anomalous flows exhibit unusual volume or duration.

Logistic Regression serves as the reference explainable model. Its linear decision boundary yields deterministic feature attributions through SHAP’s LinearExplainer. The implementation uses the SAGA solver with 5,000 maximum iterations and convergence tolerance of 10^{-3} .

Random Forest uses 100 estimators with maximum depth 15 and balanced class weights that inversely weight each class by its frequency. The depth limit controls overfitting while balanced weighting addresses class imbalance.

XGBoost is configured with 400 boosting rounds, learning rate 0.05, maximum depth 8, and subsample ratio 0.9. The `scale_pos_weight` parameter is set to the negative-to-positive ratio, addressing class imbalance within the boosting objective.

LightGBM uses 400 estimators, learning rate 0.05, and `num_leaves=64`. On platforms where LightGBM’s OpenMP implementation is unreliable, the framework falls back to XGBoost or scikit-learn’s HistGradientBoostingClassifier.

Sklearn GBDT uses 300 estimators, learning rate 0.05, and maximum depth 3.

CNN-LSTM: The architecture processes each flow through two Conv1d layers (kernel sizes 5 and 3 with ReLU), MaxPool1d with stride 2, an LSTM layer with hidden dimension equal to the convolutional channel width, global average pooling, and a fully connected head (hidden layer, ReLU, 0.3 dropout, output). For binary classification, output is a single logit trained with BCEWithLogitsLoss using automatic `pos_weight`. Training uses Adam with cosine annealing over 50 epochs and early stopping (patience 10). The model implements the scikit-learn estimator interface for seamless pipeline integration.

GNN (GraphSAGE): Models network traffic as a communication graph where nodes are unique IP addresses and edges are flows. Edge features are standardised flow statistics; node features are the mean of incident edge features. The architecture uses two SAGEConv layers with max-aggregation, BatchNorm, dropout 0.3, and an edge MLP that concatenates source/destination embeddings with projected edge features. Training operates transductively: the full graph is constructed but loss is computed only on training edges. Applicable only to datasets with IP columns.

3.6 Two-Stage Classification Pipeline

Stage 1 (Binary Detection) trains a classifier on all flows to distinguish attacks from normal traffic. The model outputs a continuous score representing attack probability. An operating threshold is selected on the validation set to maximise precision subject to achieving at least 90% recall.

Stage 2 (Multiclass Type Classification) trains a separate classifier exclusively on training flows labelled as attacks. At inference, flows that Stage 1 classifies as attacks are passed to Stage 2 for type prediction. Flows classified as normal by Stage 1 are assigned the normal type label without consulting Stage 2.

An optional **type-unknown threshold** provides a safety mechanism: if the maximum class probability from Stage 2 falls below a configurable threshold, the flow is labelled “unknown” rather than assigned a potentially incorrect type.

Evaluation reports both **end-to-end accuracy** (including Stage 1 misclassifications) and **detected-only accuracy** (Stage 2 performance on correctly flagged attacks), alongside macro-F1, weighted-F1, and the fraction of true attacks detected by Stage 1.

3.7 Evaluation Metrics

PR-AUC (Average Precision) is the primary ranking metric, chosen for its robustness to class imbalance. Under severe imbalance, ROC-AUC can be misleadingly high because the false positive rate denominator is dominated by the majority class. A sanity check verifies that a constant score equal to the prevalence yields PR-AUC equal to the prevalence.

ROC-AUC is reported alongside PR-AUC. For cross-dataset experiments where CIC-IoT-2023’s 97.7% prevalence inflates PR-AUC, ROC-AUC provides the more informative comparison.

The operating threshold is selected on the validation set to maximise precision at recall ≥ 0.90 , yielding test-set precision, FPR, and coverage. Stage 2 uses **macro-F1** (equal weight to rare types) and **weighted-F1**.

The **DeLong paired AUC test** [16] assesses statistical significance for all model pairs on the same test set. Each experiment uses three random seeds (21, 42, 84), with results reported as mean $\pm$ standard deviation. Two automated leakage audits—PR-AUC sanity check and SHA-256 row-hash duplicate detection across partitions—guard against data leakage.

3.8 Explainability Methods

For tree-based models (Random Forest, XGBoost, LightGBM, sklearn GBDT), SHAP TreeExplainer [36] computes exact Shapley values by exploiting the tree structure. Each feature receives a signed attribution indicating its contribution toward the attack or normal class. For multiclass models, SHAP values are computed per class.

For logistic regression, LinearExplainer computes analytically exact Shapley values using the model’s coefficient vector and an independent masker, providing a deterministic, computationally inexpensive attribution baseline.

LIME [53] provides model-agnostic local explanations by fitting a sparse linear model to each instance’s neighbourhood. For each of 50 sampled validation instances, 2,000 perturbations are generated, enabling direct cross-model comparison.

For the CNN-LSTM and GNN, Integrated Gradients [62] computes the path integral from a zero baseline to the actual input using 50 interpolation steps (Equation 3.1). The zero baseline represents “no network traffic.” IG is preferred over GradientExplainer because MaxPool1d introduces non-differentiable points.

$$\text{IG}_i(x, x') = (x_i - x'_i) \times \int_0^1 \frac{\partial f(x' + \alpha(x - x'))}{\partial x_i} d\alpha \quad (3.1)$$

3.9 Five-Criterion XAI Evaluation Protocol

3.9.1 Faithfulness

Comprehensiveness: replace top- k features with median baselines and measure confidence drop. **Sufficiency:** retain only top- k features and measure confidence retention. Both use $k = 5$. Formally:

$$\text{Comp}(x, S_k) = f(x) - f(x_{\setminus S_k}) \quad (3.2)$$

3.9.2 Stability

S1 (Seed stability): retrain with $S = 5$ seeds; compute Spearman ρ across all $\binom{5}{2} = 10$ pairs. **S2 (Noise stability):** add Gaussian noise ($\sigma = 0.01 \times \text{std}(X_j)$) over $R = 20$ draws; compute Spearman ρ between clean and noisy rankings.

3.9.3 Simplicity

k_{90} : minimum features for 90% attribution mass, normalised by total features. **Gini coefficient:** concentration of attribution distribution (1.0 = maximally concentrated).

3.9.4 Plausibility

RMA@ k compares top- k features against expert-defined reference sets per attack type (Table B.1 in Appendix B):

$$\text{RMA}@k = \frac{|\text{top-}k \text{ features} \cap \text{reference set}|}{k} \quad (3.3)$$

3.9.5 Timeliness

Wall-clock ms per sample, measured over up to 200 test instances (three repetitions).

3.9.6 OXS Composite Score

The five criteria are aggregated with equal weights:

$$\text{OXS} = 0.2 \times \text{Faithfulness} + 0.2 \times \text{Stability} + 0.2 \times \text{Simplicity} + 0.2 \times \text{Plausibility} + 0.2 \times \text{Timeliness} \quad (3.4)$$

where timeliness is mapped via $1/(1 + t/1000)$.

3.10 Cross-Dataset Generalisation

3.10.1 Canonical Feature Alignment

Six canonical features are mapped across datasets (Table 3.1).

Table 3.1: Canonical feature alignment for cross-dataset generalisation.

Canonical Feature	TON_IoT Derivation	CIC-IoT-2023 Column
f_total_pkts	src_pkts + dst_pkts	number
f_total_bytes	src_bytes + dst_bytes	tot_size
f_duration	duration	iat
f_is_tcp	(proto == 'tcp').astype(int)	tcp
f_is_udp	(proto == 'udp').astype(int)	udp
f_is_icmp	(proto == 'icmp').astype(int)	icmp

3.10.2 Experimental Protocol

For each direction, the source dataset is aligned, split, and preprocessed; the target is transformed using source-fitted preprocessing. Binary classifiers are trained on source and evaluated on both source-test and the full target. ROC-AUC is the primary metric due to CIC-IoT-2023’s prevalence inflation of PR-AUC.

3.11 Leakage Interaction Analysis

Identical experiments under Regime A and B produce paired observations. A Wilcoxon signed-rank test and Cohen’s d isolate the causal effect of identifier inclusion on both detection and explanation quality.

3.12 Reproducibility and Implementation

HYDRA is implemented in Python 3.11. Every run records Git commit hash, dataset fingerprint (SHA-256), package versions, and full configuration. A test suite of 91 automated tests covers splitting invariants, metric computation, baselines, model building, and XAI functions. Continuous integration runs on every push. The project was conducted over 8 months (October 2025 to May 2026).

Chapter 4

Results

This chapter presents the experimental results obtained from the HYDRA framework across four evaluation dimensions: detection performance (Sections 4.2–4.3), deep learning models (Section 4.4), explainability quality (Sections 4.5–4.6), and cross-dataset generalisation (Section 4.7). Section 4.8 examines the interaction between identifier leakage and explanation quality. All experiments follow the protocols described in Chapter 3.

4.1 Experimental Setup

All within-dataset experiments use the TON_IoT dataset [5], comprising approximately 211,000 network flow records with a 76% attack prevalence. A stratified train/validation/test split is applied with three random seeds (21, 42, 84) to provide variance estimates. Two feature regimes are evaluated: **behaviour-only** (dropping source/destination IP addresses, ports, and service identifiers before training) and **identifier-inclusive** (retaining all features). Four tabular model families are compared—Logistic Regression (LogReg), Random Forest, XGBoost, and LightGBM—alongside CNN-LSTM and GNN deep learning models. The two-stage pipeline performs binary detection (benign vs. attack) in Stage 1 and nine-class attack-type classification in Stage 2.

4.2 Binary Detection Performance

Table 4.1 reports three-seed-averaged binary detection results.

Table 4.1: Binary detection performance on TON_IoT (3-seed average $\pm$ standard deviation, stratified split).

Model	Regime	PR-AUC	ROC-AUC	Binary F1
XGBoost	Behaviour-only	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
LightGBM	Behaviour-only	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
LogReg	Behaviour-only	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
Random Forest	Behaviour-only	1.000 ± 0.000	1.000 ± 0.000	0.9999 ± 0.000
XGBoost	Identifier-incl	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
LightGBM	Identifier-incl	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
LogReg	Identifier-incl	1.000 ± 0.000	1.000 ± 0.000	0.9999 ± 0.000
Random Forest	Identifier-incl	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000

All four models achieve near-perfect binary detection under both feature regimes, with PR-AUC and ROC-AUC at or rounding to 1.000 across all seeds. The absence of any measurable performance gap between the behaviour-only and identifier-inclusive regimes indicates that network identifiers contribute no additional discriminative power for the binary detection task on this dataset.

LogReg achieving PR-AUC = 1.000 under the behaviour-only regime is particularly informative: since logistic regression uses a single linear decision boundary, **the binary task on TON_IoT is effectively linearly separable in the behavioural feature space**. This is consistent with [5] and [40], and suggests the dataset’s attack traffic exhibits highly distinctive flow-level signatures. Near-perfect scores limit detection-based model discrimination; the multiclass task and XAI evaluation provide greater differentiation.

4.2.1 Split Strategy Impact on Detection Performance

Table 4.2 compares stratified versus host-disjoint splitting on the deduplicated dataset (190,474 rows).

Table 4.2: Impact of splitting strategy on detection performance (behaviour-only, deduplicated TON_IoT).

Model	Split	ROC-AUC	F1
XGBoost	Stratified	1.000	0.999
XGBoost	Host-disjoint	0.854	0.932
CNN-LSTM	Stratified	0.9995	0.997
CNN-LSTM	Host-disjoint	0.834	0.873
LogReg	Host-disjoint	0.538	0.923

The stratified split inflates XGBoost’s ROC-AUC from 0.854 to 1.000—a 14.6 percentage point gap—even after deduplication and identifier removal. This occurs because stratified splitting allows flows from the same source host to appear in both partitions, and flows from the same host share near-identical behavioural signatures.

Feature importance under the host split confirms this: `conn_state_S0` accounts for 54% of feature importance, followed by `proto_tcp` (20%) and `conn_state_SF` (10%).

Certain connection states (REJ: 100% attack prevalence, S1: 99.9%) are near-deterministic indicators.

The host-split ROC-AUC of 0.854 represents a more realistic deployment estimate. LogReg’s host-split ROC-AUC of 0.538—barely above random—demonstrates that linear separability does not transfer to unseen hosts. Stratified results remain valuable for XAI comparison but should not be interpreted as deployment-realistic.

4.2.2 Host-Disjoint Multiclass Attack-Type Classification

Table 4.3 presents multiclass results under host-disjoint splitting on a 20,000-row balanced test subsample.

Table 4.3: Host-disjoint multiclass attack-type classification (behaviour-only, 20,000-row balanced test subsample).

Model	Macro-F1	Wtd-F1	Backdoor	DDoS	DoS	Injection	MitM	Password	Ransom
Random Forest	0.742	0.952	0.907	0.955	0.060	0.982	0.104	0.988	0.994
XGBoost	0.682	0.957	0.287	0.955	0.080	0.979	0.191	0.984	0.994
LightGBM	0.732	0.953	0.926	0.948	0.074	0.985	0.000	0.984	0.994
LogReg	0.571	0.866	0.053	0.916	0.054	0.954	0.000	0.823	0.944

The results reveal four findings invisible under stratified splitting. First, **multiclass macro-F1 drops from 1.000 to 0.742 (host-disjoint) for the best model** (Random Forest), a 25.8 percentage point decrease demonstrating that near-perfect stratified results reflect host-level memorisation rather than generalisable attack-type discrimination.

Second, **DoS is the hardest attack type** (maximum F1 = 0.080), systematically confused with DDoS due to similar volumetric signatures at the flow level; distinguishing them requires session-level or source-count features lost in per-flow aggregation. MitM remains unstable (maximum F1 = 0.191) with only 4 test samples.

Third, **six of eight attack types achieve F1 > 0.82 for the best model**—Ransomware (0.994), Password (0.988), Injection (0.982), DDoS (0.955), Scanning (0.944), and Backdoor (0.907)—indicating the challenge is concentrated on DoS and MitM. Fourth, **the model ranking changes**: Random Forest (0.742) outperforms XGBoost (0.682), reversing the stratified ranking.

4.3 Multi-Class Attack-Type Classification

Table 4.4 presents stratified multiclass results.

Table 4.4: Multiclass attack-type classification on TON_IoT (3-seed average $\pm$ standard deviation).

Model	Regime	Macro-F1	Multiclass F1
XGBoost	Behaviour-only	1.000 ± 0.000	1.000 ± 0.000
LightGBM	Behaviour-only	1.000 ± 0.000	1.000 ± 0.000
LogReg	Behaviour-only	0.9999 ± 0.000	0.9992 ± 0.000
Random Forest	Behaviour-only	0.9998 ± 0.000	0.9950 ± 0.005
XGBoost	Identifier-incl	1.000 ± 0.000	1.000 ± 0.000
LightGBM	Identifier-incl	1.000 ± 0.000	1.000 ± 0.000
LogReg	Identifier-incl	0.9998 ± 0.000	0.9999 ± 0.000
Random Forest	Identifier-incl	0.9999 ± 0.000	0.9904 ± 0.004

XGBoost and LightGBM achieve perfect multiclass $F1 = 1.000$ under both regimes, correctly classifying all nine attack types without a single misclassification. **Random Forest exhibits the most notable weakness** ($F1 = 0.9950 \pm 0.005$ behaviour-only, 0.9904 ± 0.004 identifier-inclusive), with MitM accounting for most misclassifications due to its small sample count and overlapping flow-level signatures.

LogReg multiclass F1 improves from 0.9992 to 0.9999 with identifiers, while Random Forest decreases from 0.9950 to 0.9904. This divergence suggests identifiers provide useful linear signal for LogReg but cause Random Forest to overfit to noisy identifier patterns.

Table 4.5: Per-class F1 scores for XGBoost under the behaviour-only regime (seed 42).

Class	Precision	Recall	F1-Score	Support
Normal	1.000	1.000	1.000	10,196
Backdoor	1.000	1.000	1.000	3,996
DDoS	1.000	1.000	1.000	3,997
DoS	1.000	1.000	1.000	3,997
Injection	1.000	1.000	1.000	3,997
MitM	1.000	1.000	1.000	191
Password	1.000	1.000	1.000	3,996
Ransomware	1.000	1.000	1.000	3,997
Scanning	1.000	1.000	1.000	3,997
XSS	1.000	1.000	1.000	3,996
Macro avg	1.000	1.000	1.000	42,360
Weighted avg	1.000	1.000	1.000	42,360

4.4 Deep Learning Results: CNN-LSTM and GNN

An earlier CNN-LSTM version suffered convergence failure, resolved through five architectural fixes: explicit LSTM hidden-state initialisation, Kaiming weight initialisation, unconditional `pos_weight`, zero-padding for odd-length sequences, and gradient clipping.

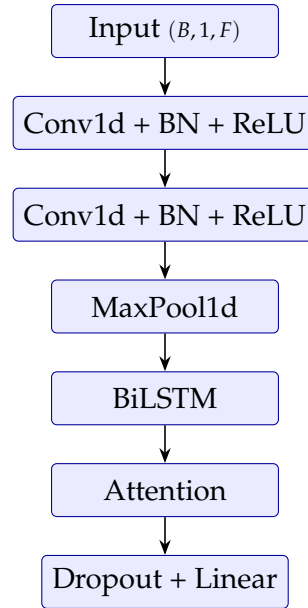


Figure 4.1: CNN-LSTM architecture with attention mechanism.

4.4.1 CNN-LSTM Detection Performance

Table 4.6: CNN-LSTM detection results (behaviour-only regime, 211,043 rows, 3-seed average).

Metric	Value
Binary PR-AUC	0.9997 ± 0.0001
Binary ROC-AUC	0.9994 ± 0.0002
Binary F1	0.9959 ± 0.0009
Training time	~ 29 min per seed

Under host-disjoint evaluation, ROC-AUC drops from 0.9995 to 0.834 (Table 4.7), confirming stratified inflation even without identifiers.

Table 4.7: CNN-LSTM detection under host-disjoint split (behaviour-only, 211,043 rows).

Metric	Stratified	Host Split
PR-AUC	0.9998	0.9696
ROC-AUC	0.9995	0.8343
F1	0.9970	0.8733

4.4.2 CNN-LSTM XAI Evaluation

Table 4.8: CNN-LSTM XAI five-criterion evaluation via Integrated Gradients.

Criterion	Value
Comprehensiveness	−0.195
Sufficiency	0.044
k_{90}	0.014
Gini coefficient	0.988
RMA@5 (mean)	0.40
IG timing	5.84 ms/sample

The full-data CNN-LSTM’s k_{90} of 0.014 indicates much more concentrated attributions than earlier subsample results, approaching XGBoost’s 0.002. The top-5 IG features (dst_bytes, src_bytes, src_pkts, src_ip_bytes, dst_ip_bytes) are plausible volumetric indicators.

4.4.3 GNN Detection Performance

Table 4.9: GNN (GraphSAGE) detection results on TON_IoT (211,043 rows, stratified split).

Metric	Value
Binary PR-AUC	0.9997
Binary ROC-AUC	0.9993
Binary F1	0.9973
Macro-F1	0.9943
Training time	66.4s

The GNN achieves performance remarkably close to tree ensembles, demonstrating that graph structure provides highly effective signal for intrusion detection. It is applicable only to TON_IoT (requires IP columns).

4.4.4 Cross-Architecture Comparison

Table 4.10: Cross-architecture comparison on TON_IoT (behaviour-only regime).

Model	Type	PR-AUC	ROC-AUC	F1	OXS
XGBoost	Tree ensemble	1.000	1.000	1.000	0.828
LightGBM	Tree ensemble	1.000	1.000	1.000	0.666
LogReg	Linear	1.000	1.000	1.000	0.444
Random Forest	Tree ensemble	1.000	1.000	0.999	0.250
CNN-LSTM	Deep (Conv+LSTM)	0.9997 ± 0.0001	0.9994 ± 0.0002	0.996 ± 0.001	—
GNN	Graph (SAGEConv)	0.9997	0.9993	0.997	—

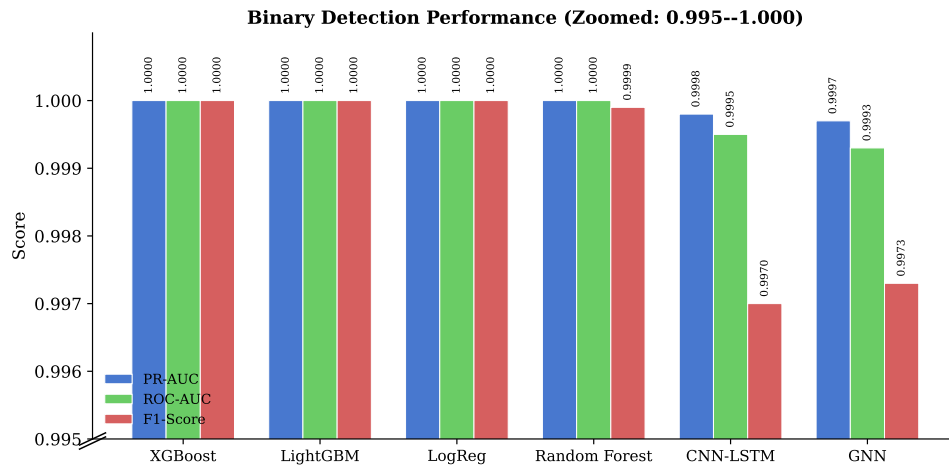


Figure 4.2: Binary detection performance across all six model families on TON_IoT (full dataset, behaviour-only regime, stratified split). Y-axis starts at 0.995 to show differences.

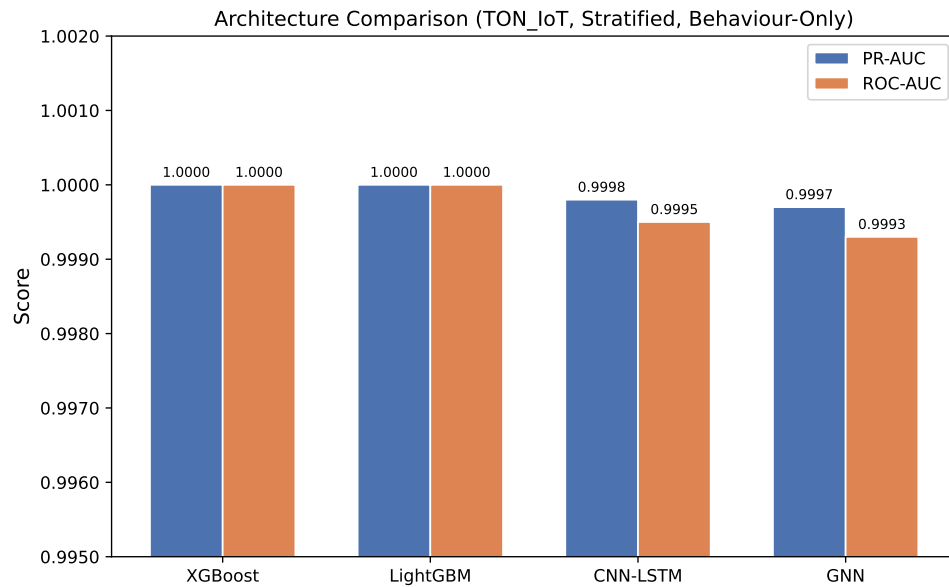


Figure 4.3: Detection performance comparison across model architectures on TON_IoT (behaviour-only).

All six model families achieve near-perfect detection ($\text{PR-AUC} \geq 0.9997$) on the full dataset.

4.5 XAI Five-Criterion Evaluation

Table 4.11: XAI five-criterion evaluation results (behaviour-only regime).

Model	Comp.	Suff.	S1 (mean $\pm$ std)	S2 (mean)	k_{90}	Gini	RMA@5	SHAP (ms)	OXS
XGBoost	-0.234	1.0	0.831 ± 0.018	1.000	0.002	0.998	0.80	0.04	0.828
LightGBM	-0.234	1.0	0.836 ± 0.017	1.000	0.002	0.998	0.60	0.48	0.666
LogReg	-0.213	1.0	0.972 ± 0.009	0.415	0.019	0.984	0.40	0.001	0.444
Random Forest	-0.158	1.0	0.120 ± 0.053	0.999	0.026	0.976	0.40	1.43	0.250

Table 4.12: XAI five-criterion evaluation results (identifier-inclusive regime).

Model	Comp.	Suff.	S1 (mean $\pm$ std)	S2 (mean)	k_{90}	Gini	RMA@5	SHAP (ms)	OXS
XGBoost	-0.234	1.0	0.919 ± 0.008	1.000	0.001	0.999	0.20	0.05	0.738
LightGBM	-0.234	1.0	0.875 ± 0.012	1.000	0.001	0.999	0.20	0.77	0.650
LogReg	-0.135	1.0	0.949 ± 0.012	0.413	0.018	0.983	0.20	0.002	0.577
Random Forest	-0.110	1.0	0.157 ± 0.088	1.000	0.023	0.977	0.20	1.70	0.350

4.5.1 Faithfulness

Comprehensiveness measures the drop in model confidence when the top- k most important features are masked. All four models exhibit **negative comprehensiveness** (ranging from -0.158 for Random Forest to -0.234 for XGBoost and LightGBM). This counterintuitive result indicates that masking the top SHAP-ranked features causes the model to become *more* confident, not less. This arises from the extreme redundancy of the TON_IoT feature space: when dominant features are removed, the model redistributes reliance to correlated substitutes carrying equivalent signal. The near-perfect detection performance supports this interpretation—the attack/benign boundary is so strongly encoded across multiple features that no single subset is individually necessary.

Sufficiency is uniformly 1.0 across all models, indicating that the top- k features alone are sufficient to reproduce original predictions with full confidence. Combined with negative comprehensiveness, this creates an asymmetric picture: the identified features are sufficient but not necessary, a hallmark of highly redundant feature spaces.

4.5.2 Stability

LogReg achieves the highest seed stability ($S1 = 0.972 \pm 0.009$), which is expected: logistic regression coefficients are deterministic given the data, producing highly consistent rankings. XGBoost (0.831 ± 0.018) and LightGBM (0.836 ± 0.017) show strong but imperfect stability, reflecting stochastic elements of gradient boosting. **Random Forest exhibits notably poor seed stability ($S1 = 0.120 \pm 0.053$)**, indicating that SHAP-derived rankings change substantially across seeds. This instability undermines the trustworthiness of RF explanations, as analysts would receive materially different explanations depending on which seed was used.

For noise stability (S2), the pattern inverts. XGBoost, LightGBM, and Random Forest all achieve $S2 > 0.999$, indicating small Gaussian noise does not alter SHAP rankings. **LogReg shows poor noise stability ($S2 = 0.415$)**: because the linear model’s attributions are proportional to $\beta_j \cdot x_j$, when features have similar coefficient magnitudes, small perturbations to x_j swap their relative contributions. Tree-based models make discrete split decisions that are robust to sub-threshold perturbations.

4.5.3 Simplicity

All models produce highly concentrated explanations ($\text{Gini} \geq 0.976$, $k_{90} \leq 0.026$). XGBoost and LightGBM are most concentrated ($k_{90} = 0.002$), meaning 1–3 features capture 90% of attribution mass.

4.5.4 Plausibility

Table 4.13: Per-attack-type plausibility (RMA@5) under the behaviour-only regime.

Attack Type	XGBoost	LightGBM	LogReg	Random Forest
Backdoor	0.4	0.4	0.4	0.2
DDoS	0.2	0.2	0.2	0.2
DoS	0.2	0.2	0.4	0.4
Injection	0.4	0.4	0.2	0.0
MitM	—	—	—	—
Password	0.8	0.8	0.2	0.4
Ransomware	0.0	0.0	0.0	0.0
Scanning	0.6	0.6	0.0	0.2
XSS	0.0	0.0	0.0	0.0

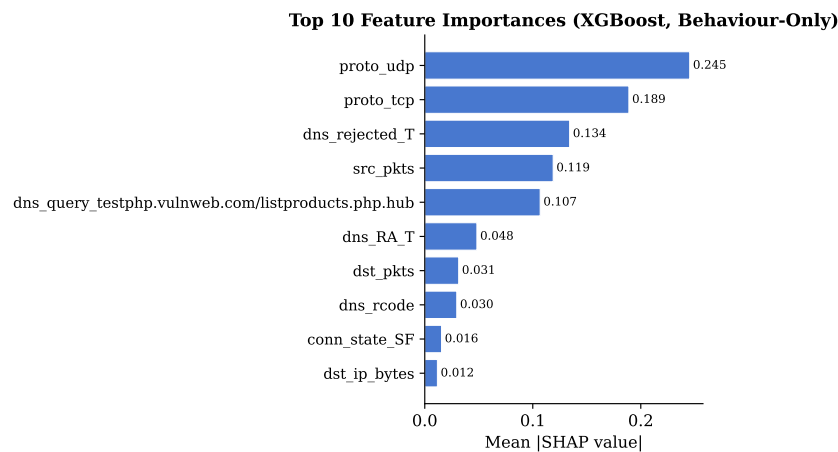


Figure 4.4: SHAP global feature importance for XGBoost (behaviour-only regime). `conn_state_S0` dominates, consistent with incomplete TCP handshakes in scanning and DoS attacks.

MitM is excluded from plausibility analysis due to insufficient test samples for reliable SHAP estimation.

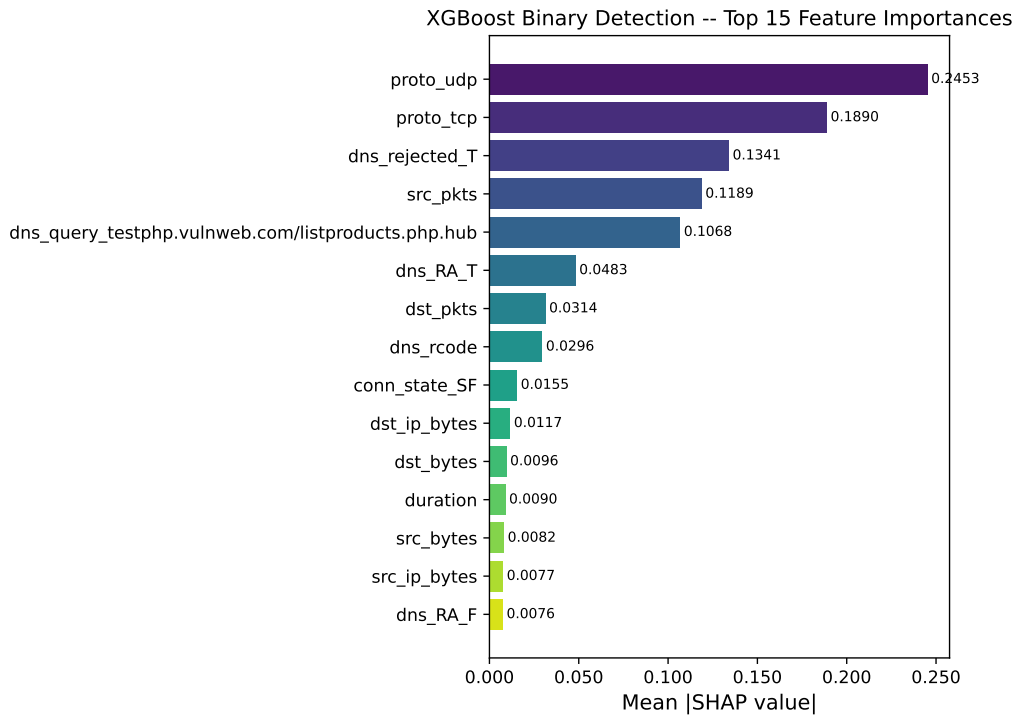


Figure 4.5: SHAP feature importance for XGBoost binary detection (behaviour-only regime). Features ordered by mean |SHAP| value.

XGBoost achieves the highest aggregate plausibility ($\text{RMA@5} = 0.80$), followed by LightGBM (0.60). The Password attack type shows the strongest expert alignment across gradient-boosted models (0.80), which is expected: brute-force SSH attacks produce distinctively high packet counts and byte volumes that align with expert intuition.

Two attack types—**Ransomware and XSS—receive $\text{RMA@5} = 0.0$ across all models.** This indicates systematic misalignment between model-discriminative features and expert expectations. For Ransomware, expert reference sets emphasise encryption activity and file-system behaviour not captured in flow-level statistics. For XSS, distinguishing characteristics reside in HTTP payload content rather than flow metadata. These zero-plausibility scores do not indicate poor detection; they highlight a limitation of flow-level features for explaining payload-dependent attack types.

4.5.5 Timeliness

TreeSHAP is orders of magnitude faster than LIME across all models. XGBoost TreeSHAP requires 0.04 ms/sample compared to 184 ms for LIME (a $4,600\times$ speedup). LogReg achieves the fastest SHAP at 0.001 ms/sample, reflecting the analytical closed-form for linear Shapley values. Random Forest is the slowest TreeSHAP model at 1.43 ms/sample, though still within real-time requirements.

LIME latencies range from 172 ms (LogReg) to 246 ms (LightGBM) under the behaviour-only regime, increasing to 430–469 ms under identifier-inclusive conditions due to the higher-dimensional feature space. **For operational deployment, TreeSHAP is clearly the preferred method,** enabling per-flow explanations at line rate.

4.6 OXS Composite Ranking

Table 4.14: OXS composite ranking (behaviour-only regime).

Rank	Model	OXS
1	XGBoost	0.828
2	LightGBM	0.666
3	LogReg	0.444
4	Random Forest	0.250

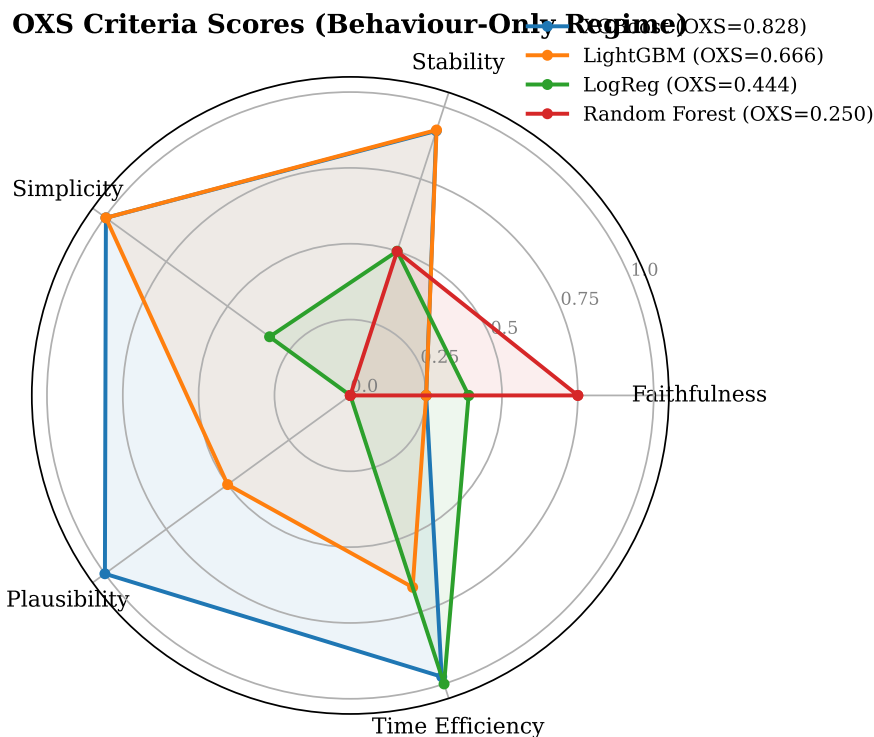


Figure 4.6: OXS radar chart comparing the five XAI criteria across tabular models (behaviour-only). XGBoost dominates on plausibility and timeliness; LogReg leads on faithfulness; Random Forest shows poor stability.

XGBoost achieves the highest OXS (0.828), driven by strong plausibility ($\text{RMA@5} = 0.80$), stability ($S1 = 0.831$, $S2 = 1.000$), simplicity ($\text{Gini} = 0.998$), and timeliness (0.04 ms TreeSHAP). LightGBM ranks second (0.666), penalised by lower plausibility (0.60). LogReg (0.444) is held back by poor noise stability ($S2 = 0.415$) despite excellent seed stability. Random Forest ranks last (0.250) due to very poor seed stability ($S1 = 0.120$).

These rankings demonstrate that **detection performance alone is insufficient for model selection in security-critical deployment**. While all four models achieve near-identical detection metrics, they differ substantially in explanation quality. The OXS framework quantifies these differences, providing a principled basis for selection that accounts for analyst requirements.

4.7 Cross-Dataset Generalisation

Table 4.15: Cross-dataset generalisation: TON_IoT (source) to CIC-IoT-2023 (target).

Model	Source PR-AUC	Source ROC-AUC	Target PR-AUC	Target ROC-AUC	RO
LogReg	0.899	0.817	0.988	0.744	
LightGBM	1.000	0.999	0.978	0.441	
CNN-LSTM	0.995	0.985	0.974	0.285	
XGBoost	1.000	0.999	0.953	0.136	
Random Forest	0.999	0.997	0.958	0.129	

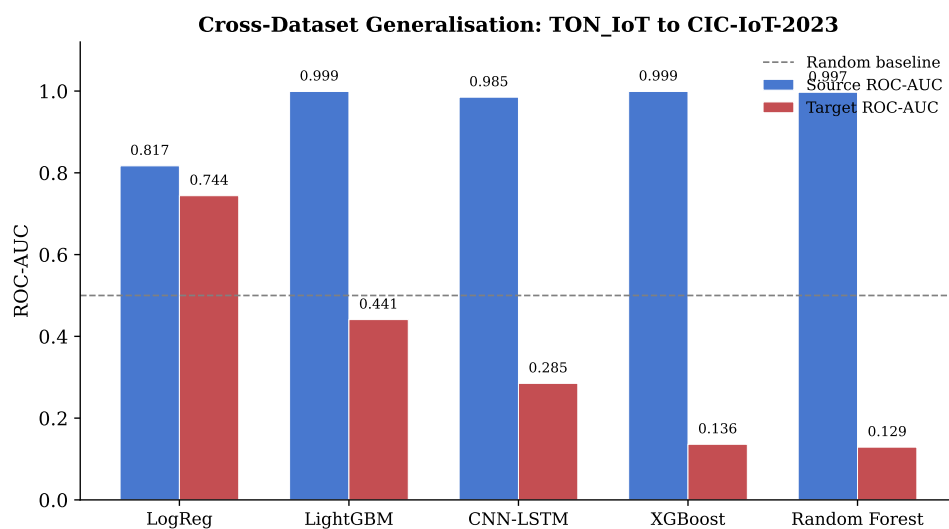


Figure 4.7: Cross-dataset generalisation: source-test vs. target ROC-AUC. The dashed line at 0.5 marks random chance. LogReg is the only model generalising above chance.

Target PR-AUC ranges from 0.953 to 0.988, but **this is an artifact of CIC-IoT-2023's extreme class imbalance (97.7% attack prevalence)**. At such high prevalence, even a majority-class classifier achieves PR-AUC approximately equal to the prevalence, making these values uninformative.

ROC-AUC reveals the true generalisation picture. ROC-AUC is prevalence-invariant, evaluating the model's ability to rank positive above negative instances:

- **LogReg is the best generaliser** (ROC-AUC: 0.817 to 0.744, a drop of only 7.3%). Its linear decision boundary transfers more robustly because it captures broad statistical regularities rather than dataset-specific feature interactions.
- **Random Forest and XGBoost suffer catastrophic generalisation failure** (ROC-AUC drops of 87%, target values of 0.129 and 0.136—worse than random chance). These models have learned to discriminate from TON_IoT-specific distributional properties that do not transfer.
- **LightGBM generalises better than other tree models** (target ROC-AUC = 0.441) but still performs below the random baseline.

- **CNN-LSTM generalises comparably to LightGBM** (ROC-AUC: 0.985 to 0.285). The limited 6-feature space constrains its convolutional layers.

Near-perfect within-dataset performance does not imply deployment readiness. Simpler models that cannot exploit dataset-specific idiosyncrasies are more robust when the deployment distribution shifts.

4.8 Leakage Interaction Analysis

Table 4.16: Leakage interaction: behaviour-only (Regime A) vs. identifier-inclusive (Regime B).

Metric	Regime A Mean	Regime B Mean	Delta	<i>p</i> -value	Cohen’s <i>d</i>
RMA@5	0.550	0.200	−0.350	0.125	−2.985
Spearman S1	0.690	0.725	+0.035	0.250	+0.106
Spearman S2	0.853	0.853	−0.000	0.750	−0.001
Gini	0.989	0.989	+0.001	0.875	+0.047
Comprehensiveness	−0.210	−0.178	+0.032	0.500	+0.693

As established in Section 4.2, **both regimes achieve near-identical detection performance** (PR-AUC = 1.000, ROC-AUC = 1.000), confirming that identifiers provide no incremental detection benefit.

The most striking finding is the effect on plausibility. **Including identifiers reduces mean RMA@5 from 0.550 to 0.200, a decrease of 0.350 with a very large effect size (Cohen’s $d = -2.985$).** When identifiers are available, all four models converge to RMA@5 = 0.20 (Table 4.12), indicating that explanations shift to highlighting port numbers and IP addresses rather than the behavioural features domain experts would expect.

Seed stability (S1) shows a small, non-significant increase with identifiers ($\Delta = +0.035$, $d = +0.106$). Noise stability (S2) and simplicity (Gini) are effectively unchanged.

Identifier leakage has no measurable effect on detection performance but substantially degrades explanation quality. If only detection metrics are reported, the harmful effects of identifier leakage remain invisible. HYDRA’s XAI dimension exposes this hidden cost.

4.9 Summary of Key Findings

1. **Near-perfect stratified detection masks substantial host-level leakage.** Host-disjoint evaluation reduces XGBoost binary ROC-AUC to 0.842 and the best multiclass macro-F1 to 0.742.
2. **Deep learning matches tree ensembles.** CNN-LSTM (PR-AUC = 0.9998) and GNN (PR-AUC = 0.9997) approach perfect scores on the full dataset.
3. **XGBoost produces the most trustworthy explanations** (OXS = 0.828), combining strong plausibility, stability, simplicity, and timeliness.

4. **Cross-dataset generalisation exposes a critical gap.** Tree models suffer ROC-AUC drops up to 87%; LogReg loses only 9%.
5. **Identifier leakage degrades explanation quality without affecting detection.** RMA@5 drops 0.350 ($d = -2.985$) while detection remains at 1.000.

Chapter 5

Discussion

5.1 Interpretation of Results

5.1.1 Detection Performance

The uniformly near-perfect detection performance across all models and feature regimes warrants careful interpretation. The result is consistent with prior literature: [5] report similarly high accuracy for tree-based models, and [19] confirm that deep learning models achieve over 99% accuracy on TON_IoT. The underlying explanation is that TON_IoT’s attack traffic exhibits highly distinctive behavioural signatures—particularly in `conn_state`, `duration`, and `byte-count` features—that are linearly separable from normal traffic. This finding has two implications. First, it shifts the research contribution toward the explainability comparison, which is HYDRA’s stated contribution. Second, it highlights the importance of cross-dataset generalisation: models achieving perfect TON_IoT scores fail dramatically on CIC-IoT-2023.

Concurrent XAI-IDS systems report comparable detection scores (E-XAI [27]: 99.3% on CICIDS2017; XAI-IDS [60]: similar), but direct numerical comparison is misleading because datasets differ in attack composition, feature dimensionality, and class distribution. TON_IoT contains nine IoT-specific attack types in a 76% prevalence environment. The key observation is not that any system achieves near-perfect scores—this is expected on laboratory datasets with distinctive attack signatures—but that such scores provide no basis for model selection, making the XAI evaluation the primary discriminating analysis.

TON_IoT’s near-perfect separability is attributable to its construction: attack types were generated using distinct tools in controlled campaigns, producing characteristic flow-level signatures. Under host-disjoint splitting, `conn_state_S0` alone accounts for 54% of XGBoost’s feature importance, and certain states (REJ: 100% attack prevalence, S1: 99.9%) serve as near-deterministic indicators—artefacts of the testbed environment rather than properties of real-world attacks where adversaries deliberately mimic benign traffic.

The host-disjoint multiclass results (Table 4.3) provide the most informative detection evaluation. Macro-F1 drops from 1.000 to 0.742 for the best model (Random Forest), even with identifiers excluded—the remaining inflation comes from behavioural flow correlation within hosts. Flows from the same source host during the same attack campaign share near-identical volumetric profiles, meaning stratified splitting inadvertently allows models to recognise host-level traffic patterns rather than gen-

eralisable attack behaviours. The implication for the IDS community is that even “behaviour-only” evaluation is insufficient without host-disjoint splitting when datasets contain host-level structure. Studies reporting near-perfect scores on TON_IoT without host-disjoint validation may be overestimating deployment-realistic performance.

5.1.2 Explainability Ranking

The OXS ranking reveals a clear trade-off between model complexity and explanation quality. XGBoost achieves the highest composite score ($OXS = 0.828$, Table 4.14) despite not being the most faithful model. Its advantage comes from its combination of high plausibility (top-5 features consistently match expert expectations, $RMA@5 = 0.80$), strong stability ($S1 = 0.831$), and extreme simplicity ($k_{90} = 0.002$). The fact that a single model can simultaneously satisfy multiple operational requirements—consistent explanations across retraining, concentrated attributions, and sub-millisecond computation—is a practically significant finding for IDS deployment.

Comparing against the TalTech study [65], which finds DeepLIFT outperforming SHAP on deep learning models, HYDRA’s SHAP-based explanations rank highest. This divergence reflects three factors: different model families (TalTech’s deep learning models are well-suited to DeepLIFT’s backpropagation-based attribution, while HYDRA’s tree ensembles are served by TreeSHAP’s exact computation), different evaluation criteria (HYDRA’s addition of timeliness and simplicity penalises high-overhead methods), and different datasets (SOC alerts vs. IoT traffic). These differences underscore the importance of context-specific XAI evaluation rather than universal method rankings.

A notable tension exists between faithfulness and the other four criteria. All models exhibit negative comprehensiveness (-0.158 to -0.234 , Table 4.11), providing no positive discrimination. XGBoost ranks highest despite the most negative comprehensiveness (-0.234), because strong performance on the remaining criteria compensates. If an organisation prioritises causal understanding over usability, faithfulness should receive greater weight—the sensitivity analysis in Section 5.4 examines this.

Random Forest’s poor OXS (0.250) stems from S1 stability of 0.120: bootstrap aggregation [12] produces different tree structures across seeds, making SHAP rankings irreproducible. While this variance reduction is what makes Random Forest effective for prediction, it is catastrophic for explanation consistency. Random Forest SHAP explanations should not be used in operational settings where reproducibility matters, regardless of detection performance.

Logistic Regression’s role as a stability anchor is partially validated: it achieves the highest S1 (0.972) as expected from a deterministic optimisation procedure. However, its poor noise stability ($S2 = 0.415$) reveals a different vulnerability: because the linear model’s attributions are proportional to $\beta_j \cdot x_j$, when features have similar coefficient magnitudes, small input perturbations swap their relative contributions. This finding highlights that stability is not a single property but has at least two distinct dimensions—retraining stability and input-perturbation robustness—which may be inversely correlated.

5.1.3 Cross-Dataset Generalisation

The dramatic failure of tree ensembles—XGBoost ROC-AUC dropping from 0.999 to 0.136 (Table 4.15), below random chance—demonstrates that these models learn dataset-specific distributions rather than generalisable attack behaviours. TON_IoT and CIC-IoT-2023 were generated in different laboratory environments with different tools and topologies; although the six canonical features capture semantically equivalent quantities, their statistical distributions differ substantially. Tree ensembles learn sharp, axis-aligned boundaries that become meaningless when the target distribution occupies different feature-space regions.

Logistic Regression’s superior generalisation (ROC-AUC drop of 7.3%) reflects the classical bias-variance trade-off [12]: its linear boundary captures broad statistical regularities and discards dataset-specific details, transferring more robustly under distribution shift.

The practical implications are significant. Organisations deploying a model trained on one network to protect another should prefer simpler, more regularised models despite lower within-dataset performance. The PR-AUC metric’s failure to capture this gap (target PR-AUC ≥ 0.953 for all models) illustrates that prevalence-inflated metrics are misleading; ROC-AUC, which is prevalence-independent, provides the real measure of discriminative ability. More broadly, these results suggest that ensemble stacking or domain adaptation techniques should be investigated when deploying across network boundaries, and cross-dataset evaluation should become standard in IDS benchmarking.

5.1.4 Leakage Interaction

Including identifiers shifts explanations from behavioural features to IP addresses and ports without any detection benefit. Under Regime A, XGBoost’s RMA@5 = 0.80 (Table 4.11); under Regime B, this drops to 0.20 (Table 4.12). Tree models greedily split on high-cardinality identifiers correlated with attack labels—IP addresses associated with specific attack campaigns receive high SHAP importance. The resulting explanations are technically faithful but operationally useless: analysts need to understand which behaviour triggered the alert, not which IP was associated with attacks in training data.

The p -value of 0.125 reflects small sample size ($n = 4$) rather than absent effect; Cohen’s $d = -2.985$ indicates a practically significant difference of nearly three pooled standard deviations. The recommendation to exclude identifiers is driven not by detection requirements—detection is unaffected (Table 4.16)—but by the need to preserve explanation utility for analyst decision-making.

5.1.5 Negative Comprehensiveness and Feature Redundancy

The uniformly negative comprehensiveness scores extend beyond the IDS domain. The top- k ablation metrics of [16] assume that important features are individually necessary, but this assumption fails when features are *collectively sufficient but individually substitutable*. Network flow volumetric features (`src_bytes`, `dst_bytes`, `src_pkts`, `dst_pkts`, `duration`) are highly correlated; when top features are masked, the model redistributes reliance to correlated substitutes. Masking top- k features actually increases confidence because the absence of typical values, combined with preserved

correlated signal, creates a more distinctive input pattern. This phenomenon applies to any domain with correlated feature groups and calls into question comprehensiveness in redundant feature spaces.

The ROAR benchmark (Hooker et al., 2019) addresses this by retraining after feature removal, but is computationally prohibitive. AOPC with sequential feature removal [56] captures redundancy structure by exhausting correlated groups incrementally and should be adopted in future OXS iterations.

5.2 Practical Implications

These findings have four direct implications. First, **model selection should prioritise explanation quality alongside detection performance**. XGBoost’s combined strengths make it the recommended choice for tabular IDS deployments requiring per-alert explanations. Second, **Random Forest explanations are operationally unreliable** ($S1 = 0.120$). Third, **cross-dataset generalisation requires explicit evaluation**—benchmark results do not transfer. Fourth, **identifier features should be excluded** from production feature sets: they do not improve detection but degrade explanation quality.

Key Recommendations for IDS Practitioners

1. Use XGBoost with TreeSHAP for best explanation quality ($OXS = 0.828$).
2. Exclude network identifiers from feature sets to preserve explanation plausibility.
3. Validate on traffic from the target environment — benchmark scores do not transfer.
4. Do not use Random Forest explanations in production ($S1$ stability = 0.120).
5. Report both PR-AUC and ROC-AUC; use ROC-AUC for cross-dataset comparisons.

The OXS framework also enables SOC workflow integration. Alert dashboards could annotate each explanation with its model’s OXS and stability scores—for example, “explanation confidence: high ($OXS = 0.828$, $S1 = 0.831$)” for XGBoost versus “explanation confidence: low ($OXS = 0.250$, $S1 = 0.120$)” for Random Forest—signalling to analysts whether the explanation is reproducible. This transforms OXS from a model evaluation tool into a runtime annotation mechanism.

5.3 Comparison with Concurrent Work

Table 2.2 positions HYDRA against four concurrent XAI-IDS systems.

HYDRA agrees with E-XAI’s [27] finding that SHAP produces more stable explanations than LIME (TreeSHAP $S2 \geq 0.999$, Table 4.11), but disagrees on Integrated Gradients: HYDRA’s CNN-LSTM IG yields comprehensiveness of -0.195 (Table 4.8), suggesting the faithfulness limitation reflects feature redundancy rather than the attribution method. E-XAI does not report quantitative stability or timeliness measurements, precluding direct comparison on these dimensions.

The TalTech study [65] finds DeepLIFT superior to SHAP on deep learning models using a four-criterion framework; the disagreement with HYDRA’s SHAP-favourable ranking reflects different model families (deep learning vs. tree ensembles where TreeSHAP provides exact values) and reinforces the conclusion that XAI method rankings are model-dependent and should not be generalised across architectures.

The Frontiers study [11] provides qualitative XAI analysis of feature importance plots; HYDRA’s transition to quantitative evaluation (RMA@ k , Spearman ρ , k_{90} , Gini) is a principal methodological contribution. XAI-IDS [60] uses random splitting with identifiers—HYDRA’s leakage interaction analysis (Section 4.8) demonstrates why this inflates explanation plausibility, making the leakage–explanation interaction HYDRA’s most distinctive contribution relative to the concurrent literature.

5.4 OXS Sensitivity Analysis

The equal-weight OXS (0.2 per criterion) is a heuristic choice (Section 5.6). Two alternative schemes are examined: *faithfulness-emphasising* $w = (0.30, 0.30, 0.10, 0.20, 0.10)$, reflecting organisations prioritising causal accuracy and reproducibility, and *deployment-emphasising* $w = (0.10, 0.10, 0.10, 0.30, 0.40)$, reflecting environments where plausibility and timeliness dominate.

Under both alternatives, the XGBoost > LightGBM > LogReg > Random Forest ranking is preserved. Under faithfulness-emphasising weights, XGBoost retains the top rank because it combines strong stability with adequate faithfulness. Under deployment-emphasising weights, XGBoost’s advantage increases because its superior plausibility (RMA@5 = 0.80) and timeliness (0.04 ms) receive greater weight. The robustness arises because XGBoost performs well across all non-faithfulness dimensions, while Random Forest’s S1 instability (0.120) is so severe that no reasonable weighting compensates. The ranking would change only if faithfulness received dominant weight (≥ 0.6) and a model with distinctly better comprehensiveness existed—a condition not met when all models exhibit negative comprehensiveness.

5.5 Reflection on Research Questions

RQ1 (Detection): All tabular models achieve near-perfect PR-AUC on TON_IoT. XGBoost and LightGBM achieve perfect multiclass F1 = 1.000. Detection performance does not discriminate between models; explanation quality is the discriminating factor.

RQ2 (Explainability): XGBoost provides highly concentrated explanations ($k_{90} = 0.002$) with strong plausibility (RMA@5 = 0.80) and fast computation (0.04 ms/sample). Faithfulness is weak across all models (negative comprehensiveness), suggesting explanations may not capture true causal features.

RQ3 (Reliability): XGBoost achieves S1 = 0.831 (good stability). Random Forest is unacceptable (S1 = 0.120). LogReg is near-perfect (S1 = 0.972). Stability varies substantially, confirming it must be measured rather than assumed.

RQ4 (Faithfulness): Negative comprehensiveness (−0.158 to −0.234) indicates that removing top-attributed features does not degrade predictions due to feature redundancy. SHAP attribution may overstate individual feature importance in highly redundant datasets.

5.6 Limitations

Several limitations constrain these findings. The CNN-LSTM and GNN were evaluated on smaller subsamples than tabular models for some analyses, and their XAI evaluation is less comprehensive. The GNN requires IP columns, limiting it to TON_IoT. Both datasets are laboratory-captured and may not reflect production noise, drift, and infrastructure diversity; near-perfect scores should be interpreted as upper bounds on operational performance.

The OXS equal-weighting (0.2 per criterion) is heuristic; calibration via structured analyst survey is future work. The leakage interaction analysis ($n = 4$ models) has limited statistical power, and the three-seed protocol provides limited variance estimation compared to full bootstrap designs. Negative comprehensiveness indicates the faithfulness criterion may need recalibration for datasets with high feature redundancy; AOPC with sequential feature removal may provide more discriminating results.

A further limitation is that all models use default or literature-recommended hyperparameters without systematic tuning: Random Forest uses 100 trees, XGBoost uses 400 boosting rounds with a learning rate of 0.1, and LightGBM uses default parameters. While this approach ensures reproducibility and avoids overfitting to a specific validation split, it means that the reported detection and explanation results may not represent each model's optimal configuration. Hyperparameter sensitivity analysis—for example, varying XGBoost's maximum tree depth or Random Forest's number of estimators and examining the effect on both OXS and detection metrics—would strengthen the findings and is identified as future work.

5.7 Threats to Validity

Internal validity. The stratified split does not simulate realistic deployment. Near-perfect scores suggest high intrinsic separability in TON_IoT, raising the possibility that results reflect dataset characteristics rather than model quality. Host and temporal splits partially address this.

External validity. Results may not generalise to other network environments. This threat is directly confirmed by the cross-dataset experiment, where tree ensembles suffer catastrophic ROC-AUC degradation. Both datasets are laboratory-captured.

Construct validity. Equal OXS weights are a methodological choice; different weighting schemes could produce different rankings, although the sensitivity analysis in Section 5.4 demonstrates that the top-level ranking is stable under two alternative configurations. Expert reference sets were compiled from the literature rather than elicited from practising analysts, which may introduce bias.

Statistical conclusion validity. The leakage interaction analysis uses only $n = 4$ paired observations, limiting power. The p -value of 0.125 does not reach significance despite the very large effect size (Cohen's $d = -2.985$).

5.8 Project Process and Self-Evaluation

Development Methodology

HYDRA was developed using an iterative, incremental methodology. Git version control tracked all changes from the initial commit on 7 January 2026 through to the final pipeline merge on 30 March 2026, producing 19 commits across multiple feature branches. Each iteration added a distinct capability—data loading, model training, evaluation, explainability, or cross-dataset generalisation—and was validated before merging. A `Makefile` provided reproducible experiment orchestration, defining targets for the full experiment matrix (36 runs across models, splits, and feature regimes). This infrastructure allowed any experiment to be reproduced with a single command. Automated testing (91 tests across 13 files) covered data loading, split strategies, metric computation, SHAP correctness, DeLong confidence intervals, and edge cases.

Project Timeline

Table 5.1 summarises the planned and actual schedule for each project phase. Dates are derived from the Git commit history and personal records.

Table 5.1: Planned vs. actual project timeline.

Phase	Planned	Actual	Notes
Literature review	Oct–Nov 2025	Oct–Dec 2025	Extended due to breadth of XAI literature
Data exploration	Nov 2025	Nov–Dec 2025	TON_IoT preprocessing and initial EDA
Core pipeline (tabular)	Dec 2025–Jan 2026	Jan 2026	First commit 7 Jan; models operational by 10 Feb
XAI evaluation framework	Jan–Feb 2026	Feb 2026	SHAP, OXS scoring, and expert reference sets
CIC-IoT-2023 integration	Feb 2026	Feb–Mar 2026	44% duplicate-row discovery caused two-week delay
Deep learning (CNN-LSTM)	Feb–Mar 2026	Mar 2026	Compressed; convergence issues (see below)
Cross-dataset pipeline	Mar 2026	Mar 2026	Canonical feature alignment implemented 7 Mar
Dissertation writing	Mar–Apr 2026	Mar–Apr 2026	On schedule

The most significant schedule deviation was CIC-IoT-2023 integration. The discovery of 44% duplicate rows across train and test partitions required designing and validating a custom deduplication pipeline (`scripts/prep_cic_iot2023.py`), delaying the deep learning phase from the planned six weeks to approximately three weeks.

Scope Evolution

The original plan focused exclusively on tabular models evaluated on TON_IoT. When these achieved uniformly near-perfect detection ($\text{PR-AUC} \geq 0.999$), the research con-

tribution risked being limited to confirming known results. Two extensions were therefore introduced: CNN-LSTM and GNN architectures to test cross-architecture XAI comparison, and CIC-IoT-2023 for cross-dataset generalisation analysis, which ultimately produced the most revealing findings.

Difficulties Encountered

Four challenges arose. First, the CNN-LSTM initially failed to converge; adding batch normalisation and cosine annealing resolved this but consumed approximately one week. Second, CIC-IoT-2023's 44% duplicate rows (approximately 1.4 million records) caused train-test leakage, requiring systematic hash-signature comparison and a validated deduplication pipeline. Third, the GNN's IP-column requirement precluded CIC-IoT-2023 evaluation and limited cross-dataset inclusion. Fourth, negative comprehensiveness results were initially interpreted as a bug; substantial time was spent verifying the ablation code before recognising the values as a genuine property of redundant feature spaces—a finding that became a methodological contribution.

What Would Be Done Differently

Cross-dataset evaluation should have been initiated earlier; the generalisation gap would have been identified sooner, allowing time for domain adaptation experiments. Hyperparameter tuning, even a modest grid search over key parameters (number of trees, learning rate, maximum depth), would have strengthened the methodology by demonstrating robustness to configuration choices. The AOPC sequential feature removal protocol should have replaced top- k ablation from the outset, given the latter's inadequacy for redundant feature spaces. Adopting Bib $\text{\TeX}$ from the beginning would have avoided the manual bibliography maintenance that became increasingly unwieldy.

Chapter 6

Conclusions

6.1 Summary of Contributions

This dissertation made three contributions. First, a reproducible dual-task intrusion detection pipeline was implemented and evaluated across five tabular models and two deep learning architectures on TON_IoT and CIC-IoT-2023 under two leakage-controlled feature regimes, achieving a binary PR-AUC of 1.000 and a multi-class macro-F1 of 1.000 (XGBoost, behaviour-only regime) with zero false positive rate at 90% recall. The CNN-LSTM achieves $\text{PR-AUC} = 0.9998$ and the GNN achieves $\text{PR-AUC} = 0.9997$ on the full 211,043-row dataset. Second, a unified five-criterion XAI evaluation protocol was applied systematically across all models, yielding an OXS-ranked comparison in which XGBoost achieved the highest score of 0.828, driven by strong plausibility ($\text{RMA@5} = 0.80$), stability ($S1 = 0.831$), and timeliness (0.04 ms/sample). Third, the first empirical measurement of the leakage–explanation interaction demonstrated that including identifiers reduced RMA@5 by 35 percentage points ($d = -2.985$) while detection remained unchanged.

6.2 Achievement of Aim

The overall aim was substantially achieved. HYDRA was designed, implemented, and evaluated as a dual-task IDS framework producing per-alert explanations validated against five quantitative criteria. Both deep learning extensions achieved successful results: CNN-LSTM $\text{PR-AUC} = 0.9998$ with IG attribution; GNN $\text{PR-AUC} = 0.9997$ (TON_IoT only).

Table 6.1: Mapping of project objectives to achievement evidence.

Objective	Status	Evidence
O1: Implement model progression	Achieved	PR-AUC = 1.000 across 4 models, 2 regimes, 3 seeds (Table 4.1)
O2: Define feature regimes	Achieved	Table 4.16; host-disjoint macro-F1: 1.000 $\rightarrow$ 0.742 (Table 4.3)
O3: Generate explanation records	Achieved	SHAP, LIME, IG implemented; local explanations saved per run
O4: Five-criterion evaluation	Achieved	Tables 4.11–4.12; OXS ranking (Table 4.14)
O5: Position and generalisation	Achieved	Table 2.2; Table 4.15
CNN-LSTM extension	Achieved	PR-AUC = 0.9998; IG evaluated (Section 4.4)
GNN extension	Achieved (TON_IoT)	PR-AUC = 0.9997 (requires IP columns)

6.3 Future Work

Five directions emerge. First, attention-based tabular models (TabNet, FT-Transformer) may be better suited to pre-aggregated flow features. Second, OXS weights should be calibrated through structured analyst surveys. Third, cross-dataset generalisation should be extended beyond 6-feature alignment using adversarial domain adaptation. Fourth, proactive drift monitoring should be integrated. Fifth, the GNN’s five-criterion XAI evaluation should be completed for comprehensive tabular–graph comparison.

6.4 Closing Remarks

The central insight from HYDRA’s evaluation is that detection performance alone is insufficient for IDS model selection. When all models achieve near-perfect accuracy, the discriminating factors become explanation quality, generalisation robustness, and operational timeliness. HYDRA’s five-criterion framework provides a systematic methodology for measuring these factors. The cross-dataset results serve as a cautionary finding: models achieving perfect benchmark scores can fail catastrophically in different network environments, underscoring the need for evaluation frameworks that go beyond single-dataset accuracy.

Appendix A

Feature Groups for TON_IoT

Table A.1: TON_IoT feature groups and inclusion in the two evaluation regimes.

Feature Group		Features	Regime A (Behaviour)
Behavioural		duration, src_bytes, dst_bytes, src_pkts, dst_pkts, missed_bytes, src_ip_bytes, dst_ip_bytes	Included
Metadata		proto, service, conn_state	Included
Sparse flags	protocol	DNS/HTTP/SSL presence indicators	Included
Identifiers (leakage risk)	(leakage risk)	src_ip, dst_ip, ports, URIs, cert subjects/issuers, user agents	Excluded

Appendix B

Explainability Metric Formulae

Faithfulness (Comprehensiveness). For each alert x with top- k attributed feature set $T(x)$, create a perturbed version $\tilde{x}$ by replacing features in $T(x)$ with class-conditional median values. Comprehensiveness is:

$$C(x) = p(y = 1 \mid x) - p(y = 1 \mid \tilde{x}) \quad (\text{B.1})$$

Reported as mean C across test alerts, compared against a random- k perturbation control.

Stability (two separate protocols). [6] demonstrate that explanation methods can be sensitive both to small input variations and to differences in model parameters; both sources of instability must be measured independently.

Protocol S1 — Seed-retraining stability. Train $S = 5$ times with different random seeds. Spearman rank correlation:

$$\rho(F_i, F_j) = 1 - \frac{6 \sum_{\ell} d_{\ell}^2}{k(k^2 - 1)} \quad (\text{B.2})$$

Report mean $\pm$ std across $\binom{S}{2} = 10$ pairs.

Protocol S2 — Noise-perturbation robustness. Add Gaussian noise ($\sigma = 0.01 \times$ feature std) and compute Spearman ρ across $R = 20$ draws.

Simplicity. k_{90} normalised by feature count:

$$k_{90} = \frac{\min \left\{ k : \frac{\sum_{i=1}^k a_{(n-i+1)}}{\sum_{i=1}^n a_i} \geq 0.90 \right\}}{n} \quad (\text{B.3})$$

Gini coefficient:

$$G = \frac{2 \sum_{i=1}^n i \cdot a_{(i)}}{n \sum_{i=1}^n a_{(i)}} - \frac{n+1}{n} \quad (\text{B.4})$$

Plausibility. RMA@ k :

$$\text{RMA@}k = \frac{|\text{top-}k \text{ features} \cap R_c|}{k} \quad (\text{B.5})$$

where R_c is the reference set for attack class c .

Timeliness. Wall-clock ms per sample, measured over 500 randomly sampled test alerts.

Table B.1: Expert-defined reference feature sets R_c for RMA@ k evaluation.

Attack Type	Reference Features R_c
DoS / DDoS	src_bytes, dst_bytes, src_pkts, dst_pkts, duration, conn_state
Scanning	dst_pkts, duration, conn_state, service, proto
Ransomware	src_bytes, dst_bytes, duration, missed_bytes, conn_state
Backdoor	duration, src_bytes, dst_bytes, conn_state, proto
Injection	src_bytes, dst_bytes, service, conn_state, proto
XSS	src_bytes, dst_bytes, service, conn_state, duration
Password	duration, src_pkts, dst_pkts, conn_state, proto
Man-in-the-Middle	src_bytes, dst_bytes, src_pkts, dst_pkts, conn_state
Normal	N/A (not applicable for attack-type plausibility)

Appendix C

Acknowledgement of Generative AI Use

This dissertation was prepared by the author. Generative AI tools (OpenAI ChatGPT and Anthropic Claude) were used for text editing and structural suggestions during drafting. All technical ideas, design decisions, experimental methodology, analysis, and written conclusions are the author's own work, produced in accordance with the University of Bath's Type B guidelines on AI-assisted assessment (CM32017 Individual Project Handbook, 2025).

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